Baldivis Secondary College

Semester One Examination, 2016

Question/Answer Booklet

MATHEMATICS
Year 10 Pathway 2

Calculator-assumed

Name: Mark Inkey

Time allowed for this section

Reading time before commencing work: 5 minutes
Working time for section: 50 minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer Booklet
Yellow Formula Sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: notes on one unfolded sheets of A4 paper, and up to three calculators approved for use in the WACE examinations

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of exam
Section Two: Calculator-assumed	9	9	50	45	100
			Total	45	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer Booklet.

3. You must be careful to confine your response to the specific question asked and to follow any instructions that are specified to a particular question.

4. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
   - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
   - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question that you are continuing to answer at the top of the page.

5. **Show all your working clearly**. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

6. It is recommended that you **do not use pencil**, except in diagrams.

7. Your Formula Sheet is **not** to be handed in with your Question/Answer Booklet.

Question 1

(1,1,1,1,2 =6 marks)

Simplify the following:

(a) $3^2 \times 3^7 =$ 3^9 ✓

(b) $4^6 \div 4^5 =$ 4 ✓

(c) $5^0 \times 2 =$ 2 ✓

(d) $(5^2)^3 =$ 5^6 ✓

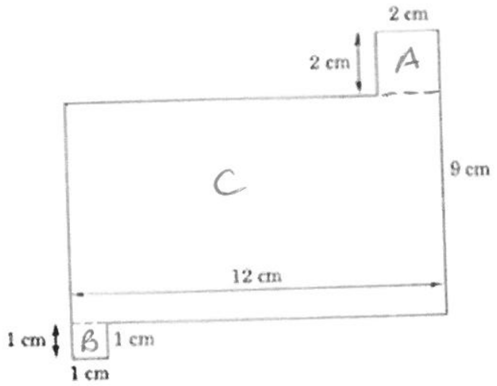
(e) $\left(\frac{u^4}{v}\right)^2$ $\frac{u^8}{v^2}$ ✓ ✓

Question 2

(4, 4 =8marks)

Find the area of these shapes:

a)



$A = 2 \times 2 = 4cm^2$ ✓

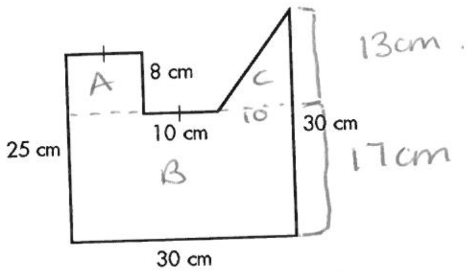
$B = 1 \times 1 = 1cm^2$ ✓

$C = 12 \times 7 = 84cm^2$ ✓

Total Area $= 84 + 1 + 4$

$= 89cm^2$ ✓

b)



$A = 10 \times 8 = 8cm^2$ ✓

$B = 17 \times 30 = 510cm^2$ ✓

$C = \frac{1}{2} \times 10 \times 13 = 65cm^2$ ✓

Total Area $= 8 + 510 + 65$

$= 583cm^2$ ✓

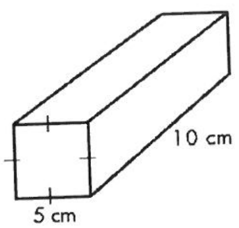
($-\frac{1}{2}$ total for no/incorrect units)

Question 3

(2, 2 = 4 marks)

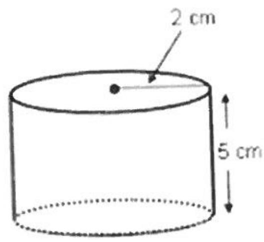
Find the volume of these solids to the nearest whole number:

(a)



$5 \times 5 \times 10 = 250cm^3$ ✓ ✓

(b)



$\pi \times 2^2 \times 5 = 62.83\,cm^3$ (2 dp) ✓ ✓

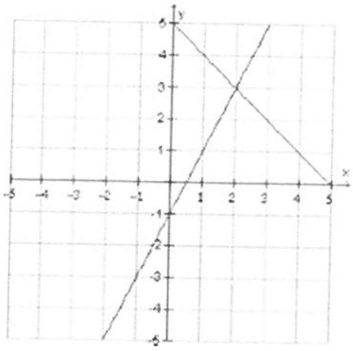
or 20π

($-\frac{1}{2}$ total for no/incorrect units).

Question 4

(2, 7, 3= 10 marks)

a) Solve this pair of simultaneous equations



$x = 2$ ✓

$y = 3$ ✓

b) Solve this pair of simultaneous equations by completing the table of values and graphing

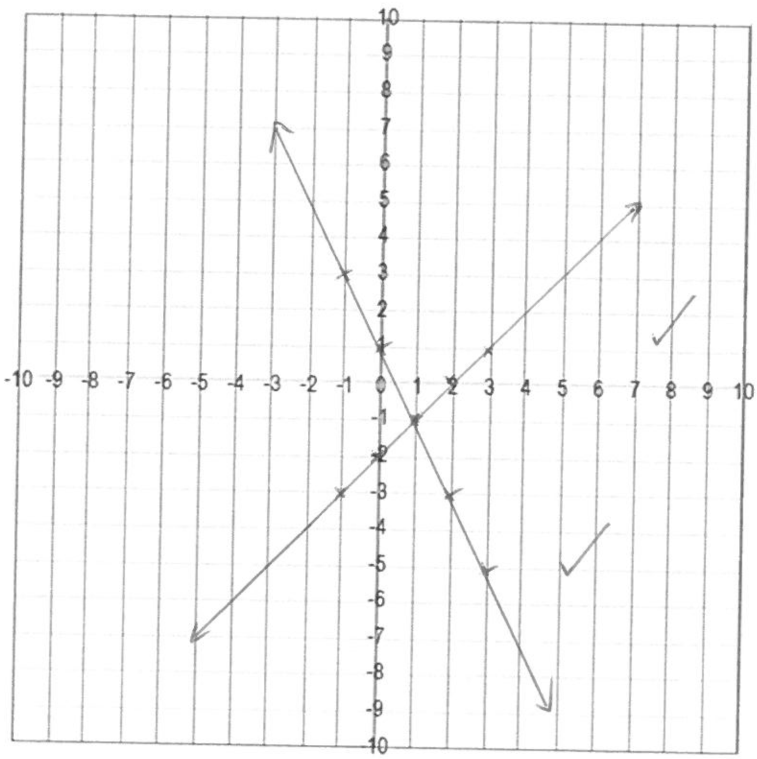
$y = x - 2$ and $y = -2x + 1$

x	-1	0	1	2	3
y	-3	-2	-1	0	1

✓✓

x	-1	0	1	2	3
y	3	1	-1	-3	-5

✓✓



$x = -1$ ✓

$y = -1$

c) Solve this pair of simultaneous equations by Substitution:

$y = 3x - 10$

$2x - 3y = 16$

$2x - 3(3x - 10) = 16$ ✓

$2x - 9x + 30 = 16$

$-7x + 30 = 16$ $(\frac{1}{2})$

$7x - 30 = -16$

$7x = 14$ ✓

$x = 2$ ✓

$y = 3(2) - 10$ $(\frac{1}{2})$

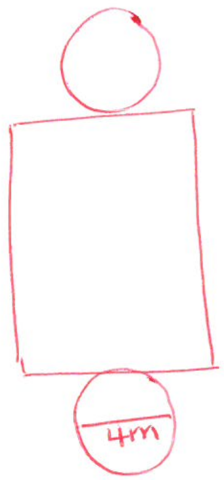
$= 6 - 10$

$= -4$ ✓

Question 5

(5)

What is the surface area of a cylinder 5 m long and 4 m in diameter? Answer correct to two decimal places.
hint: draw a diagram



Area of circle $= \pi \times 2^2$

$= 4\pi$

$\approx 12.57 m^2$ ✓

Circumference $= 2 \times \pi \times r$

$= 4\pi$

$\approx 12.57 m^2$ ✓

Curved surface $= 4\pi \times 5$

$= 20\pi$

$\approx 62.83 m^2$ ✓

Total surface area $= 2(4\pi) + 20\pi$ ✓

$= 28\pi$

$\approx 87.96 m^2$ ✓

Question 6 (1, 2, 2= 5 marks)

Expand the following

a) $3(x + 2)$ $3x + 6$ ✓

b) $(x + 3)(x - 3)$ $x^2 - 3x + 3x - 9$ ✓

$x^2 - 9$

c) $(6a - 2)^2$ $36a^2 - 12a - 12a + 4$ ✓

$36a^2 - 24a + 4$ ✓

Question 7 (1, 2, 2 =5 marks)

Factorise the following expressions:

(a) $4x + 8$ $4(x + 2)$ ✓

(b) $5x^2 + 10xy$ $5x(x + 2y)$ ✓ ✓

(c) $x^2 + 14x + 24$ $(x + 12)(x + 2)$ ✓ ✓

Question 8 (1, 1, = 2 marks)

A box contains 8 blue, 4 red and 3 white counters. Find the probability that one counter chosen is:

(a) red $\frac{4}{15}$ ✓

(b) not blue

$\frac{7}{15}$ ✓

Question 9 (2 marks)

If Alan has a $\frac{2}{3}$ chance of winning, what is his chance of losing?

$\frac{1}{3}$ ✓ ✓